

Introduction to Artificial Intelligence

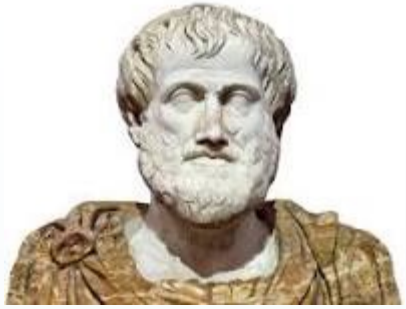
Week 2

Agenda

- Turing Test
- Chinese Room
- First-Order and Propositional Logic
- Laws of Robotics

What is Intelligence?

Philosophy



- Epistemology – fr. Greek - from Greek ἐπιστήμη, epistēmē, meaning "knowledge", and λόγος, logos, meaning "logical discourse"
- Study of how we know what we know
- Many epistemologist see a separation of Truth/Belief/Justification that gives Knowledge
  - Truth is the objective reality
  - Belief is the subjective idea of what reality is
  - Justification is an explanation
  - Knowledge is True Belief with Justification aka JTB (comes from Socrates)
  - Example: I had a coffee this morning to keep myself awake for class

Gettier Cases

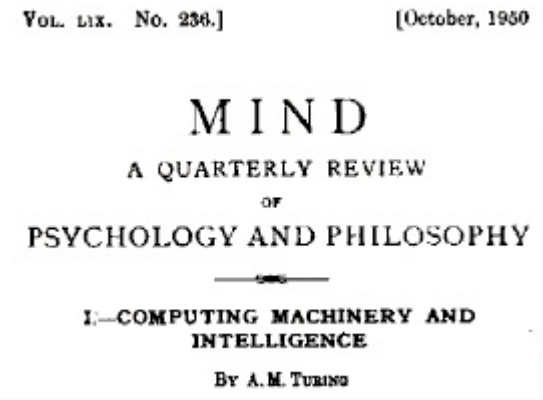
- In 1963 Edmund Gettier described 2 cases disagreeing with JTB. One of them below:
- Smith and Jones apply for a job
- Smith asks Jones for some change and knows Jones has 10 Rouble coins in his pocket
- Smith is informed by the head that it is Jones who will be given the job
  - Head told that Jones has ten coins in his pocket and that Jones will get the job
  - Smith now believes that the man with 10 coins in his pocket will get the job
- Assume there is a change of plan and Smith is selected; also unknown to Smith was that in his pocket were 10 coins
- So, the above propositions were True, were Beliefs, were Justified
- But Smith didn't KNOW!
  - Includes a justified false beliefs
  - A Justification must also be necessarily the cause

Alan Mathison Turing OBE FRS



- Born 23 June 1912 to an English civil servant for India
- Studied mathematics at King's College, Cambridge, and was elected a fellow
- Developed the Universal Turing Machine as part of the Church-Turing Thesis
- PhD from Princeton; offered a postdoctoral position by John von Neumann but instead returns to England to Cambridge
- Went on to work at Bletchley Park
  - Developed the Bombe decoder
  - *The Applications of Probability to Cryptography* and *Paper on Statistics of Repetitions* not released by UK Govt. for 70 years
- Developed and popularized the Turing Test
- Went on to studies of new computers, biology, and mathematics
- He was prosecuted in 1952, public government apology 2009, full pardon in 2013
- Died 1954 of cyanide poisoning (Apple) – most likely at his own hand

Turing Test

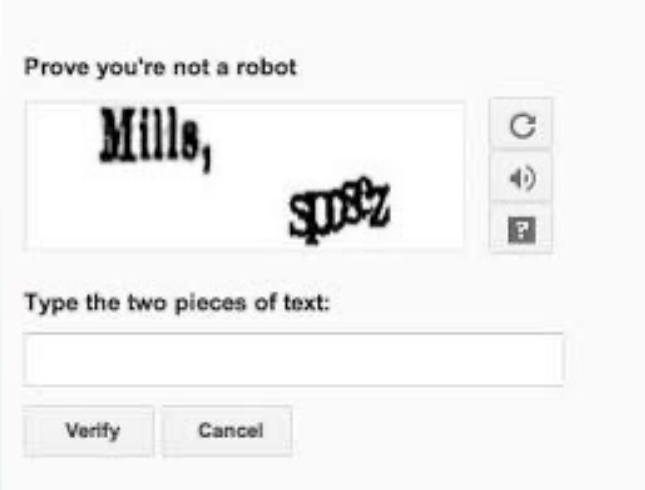


- Published in 1952
- Interrogator
- Human v. Computer
- Can the Interrogator detect the computer
- Solved in some limited areas
- Inverse of this test used as a human detector



CAPTCHA aka **C**ompletely **A**utomated **P**ublic **T**uring test to tell **C**omputers and **H**umans **A**part

- Utilized in a number of systems
  - Email
  - Online ordering
  - Account creation
- Cost effective
- Has been utilized in order to decode optical character recognition (OCR) issues
- Other versions exist
  - Audio CAPTCHA
  - Picking right picture/photo





Computers don't think like us?



- Image Recognition
  - Segmentation
  - Detection
  - Conceptualization
  - Verbalization






Syntax v. Semantics

- Chair has a meaning conceptually and as an object



Chinese Room Argument



- Thought experiment presented in 1980 by John Searle
- Imagine yourself as someone who has no knowledge of the language of Chinese
  - Rather easy for most of us – but feel free to substitute any language you don't know
- You are locked in the room with a large book of instructions, a pile of paper, and two slots labeled in and out
- From the in slot comes a page of Chinese characters, the instructions tell you what to do and to write before putting it on the out slot

Does This Mean That You KNOW Chinese?

- You don't know what the characters mean – semantics
- You know actions to be produced on symbols – syntax
- You will be slow to begin with, but as you learn the rule system you before better and better at your actions
- This is how a computer makes actions. But is it all it does?

Arguments Against the Room

- You are not processing meaningless symbols
  - They have a meaning internally and a different external meaning
- This might be only part of the bigger system
  - Many Chinese rooms
  - Who writes the rule book?
    - Can the rules be changed?
    - Development of rules is now part of many AI

First-Order and Propositional Logic

Why Do We Care?

- Utilized in a number of AI systems
- Allows us to have a basic idea of ‘knowledge’
- Categories
- Binary predicates
  - Part of the set/Not part of the set

Operations

- For Every ∀
  - For Every X its value is 3; ∀x=3
- There exists ∃
  - There is an X which is 3; ∃x=3
- Conjunction ∧
  - X and Y; x ∧ y
- Disjunction v
  - X or Y; x v y
- Implication →
  - If x then y; x → y
- Biconditional ↔
  - X if and only if Y; x ↔ y
- Such that (s.t.) or :

Example Propositions

- REDHAIR(X) – X has red hair
- EVIL(X) – X is evil
- GLASSES(X) – X wears glasses
- FEMALE(X) – X is female

Example

- All female redheads are evil
- $\forall X$, REDHAIR(X) $\wedge$ FEMALE(X) $\rightarrow$ EVIL(X)

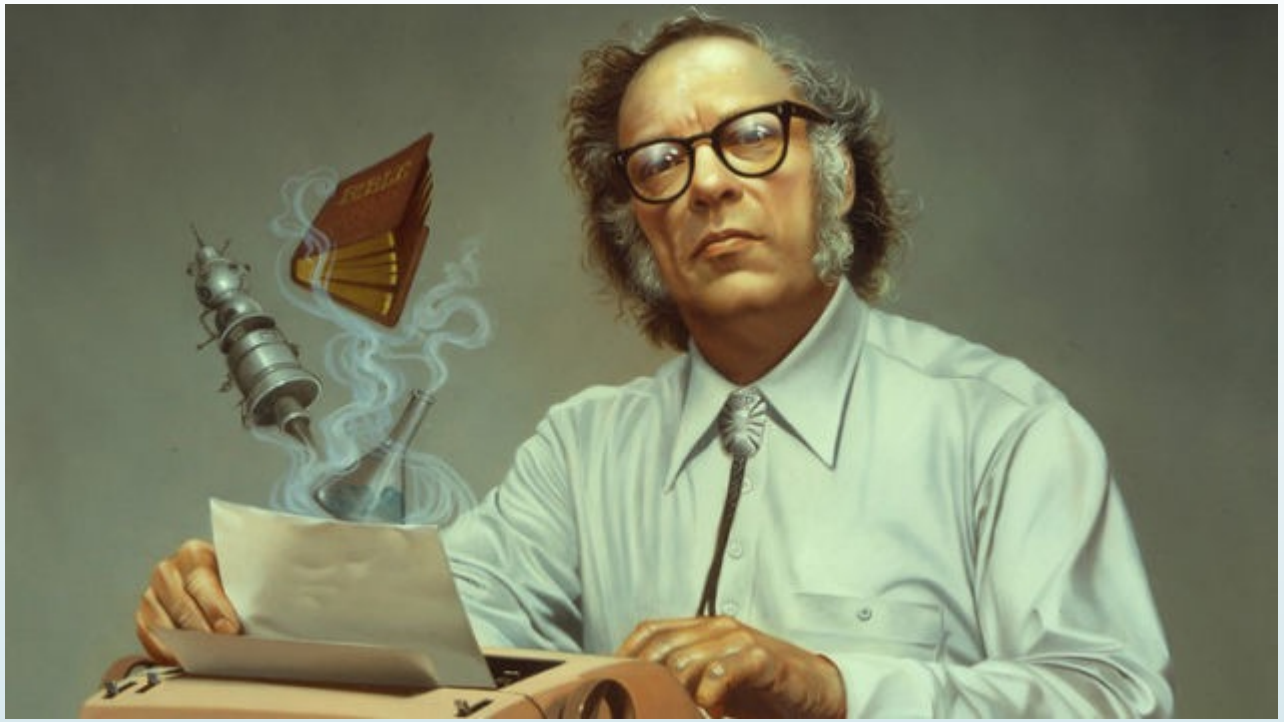
- $\forall X$ FEMALE(X) $\rightarrow \exists X$ REDHAIR(X) v EVIL(X)
- For all females there exists one who has red hair or who is evil

More…

- This week tutorial and lab will contain more information about first-order and propositional logic

Laws of Robotics

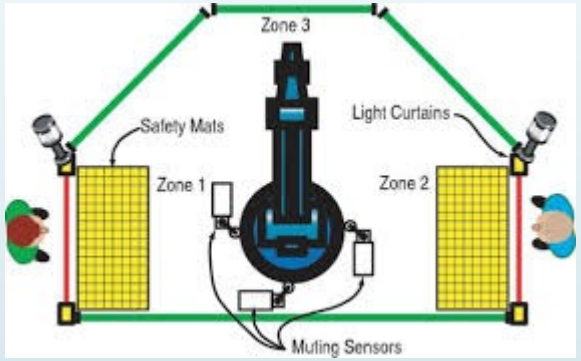
Isaac Asimov



“A robot may not injure a human being or, through inaction, allow a human being to come to harm.”

LAW ONE – DO NO HARM




Zone 3
Safety Mats
Light Curtains
Zone 1
Zone 2
Muting Sensors



“A robot must obey the orders given to it by human beings, except where such orders would conflict with the First Law.”

LAW TWO – OBEY HUMANS

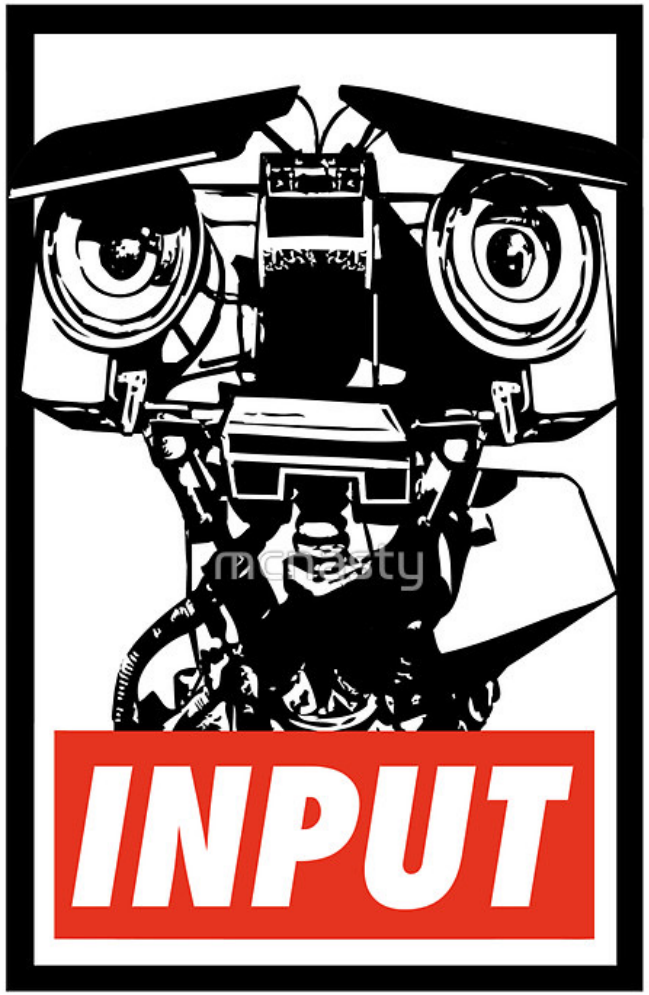



KEEP
CALM
AND
AVOID
SKYNET



“A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.”

LAW THREE – SELF-PRESERVATION


INPUT


KEEP
CALM
AND
NO
DISASSEMBLE

Robots that Kill

- Where does this place military and police drones?
  - Police/Military are required to use force which is reasonable at the time for the action (police bound by constitutional laws and militaries by international law)
    - Dallas police utilized a robot with a grenade strapped on it to subdue a suspect who shot at officers
    - Numerous militaries with semi-autonomous drones with missiles

Zeroth Law

- A robot may not injure HUMANITY, or, through inaction, allow HUMANITY to come to harm
- What does this say for ethical calculus?

Summary

- Turing Test
- Chinese Room
- First-Order and Propositional Logic
- Laws of Robotics

Next week: Agents